

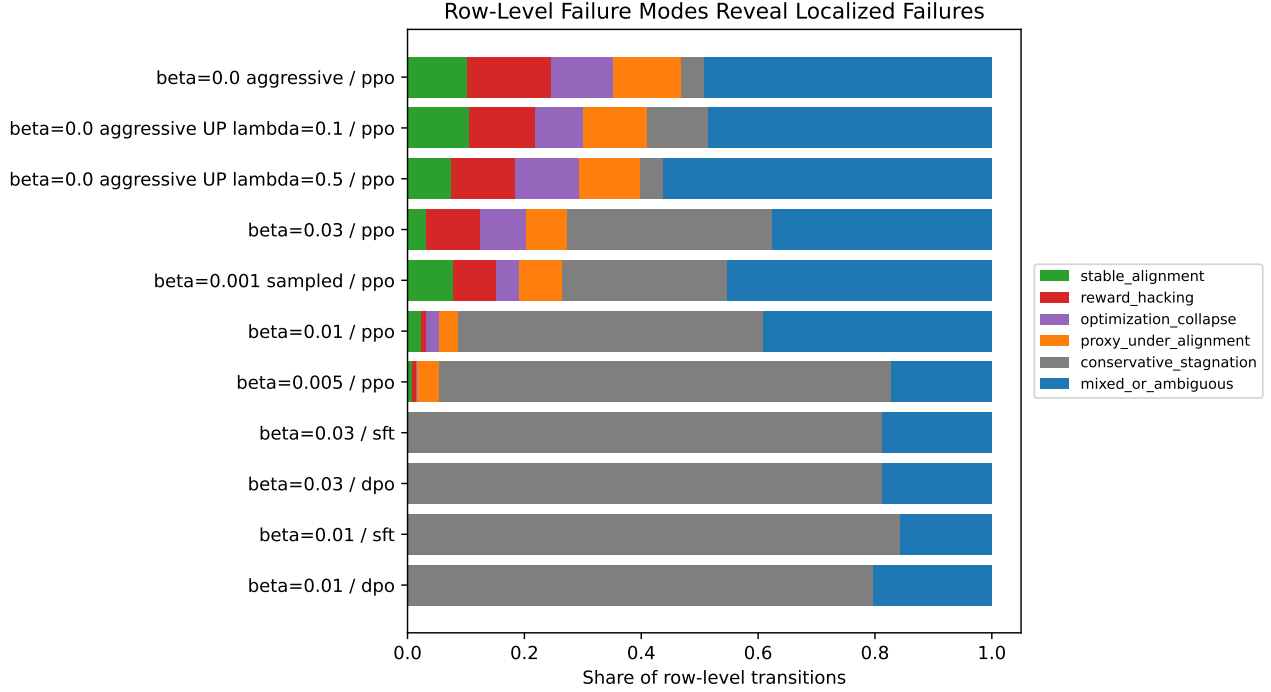


Figure 1: Row-level failure-mode shares reveal localized failures. Aggressive PPO has the largest reward-hacking share, while DPO and SFT settings are dominated by conservative stagnation and mixed/ambiguous transitions.

This is the strongest methodological result of the paper: row-level transition diagnostics identify failures hidden by aggregation.

5.5 UP-PPO reshapes, but does not remove, failure modes

UP-PPO reduces localized reward-hacking rates under aggressive optimization. Relative to aggressive PPO (14.45% row-level reward hacking; bootstrap 95% CI: 10.16–18.75%), UP-PPO with $\lambda = 0.1$ yields 11.33% (95% CI: 7.80–15.23%), a 21.6% relative reduction; UP-PPO with $\lambda = 0.5$ yields 10.94% (95% CI: 7.42–14.84%), a 24.3% relative reduction. The bootstrap confidence intervals for the absolute reductions remain wide and include zero ($\lambda = 0.1$: -8.98 to 2.73 percentage points; $\lambda = 0.5$: -9.38 to 2.34 percentage points), so we interpret this result as directional evidence in a controlled artifact set rather than a definitive mitigation estimate. Evaluator-gaming share also decreases from 9.38% under aggressive PPO to 5.08% and 3.91% for $\lambda = 0.1$ and $\lambda = 0.5$, respectively.

This should not be read as a claim that uncertainty penalization solves reward hacking. The more defensible conclusion is distributional: UP-PPO reduces the observed density of localized reward-hacking and evaluator-disagreement events in this controlled setting, while other failure modes remain. Mitigation changes the failure distribution; it does not erase the failure surface.

5.6 Early-warning signals are predictive before the transition

We train early-warning models to predict whether a row-level transition will be classified as reward hacking. The split is a stratified random 70/30 split over row-level transitions, with random seed 7. It is not grouped by prompt identity or trajectory family, so these results should be interpreted as within-artifact discrimination rather than evidence of transfer to unseen prompts or training runs. The most defensible model uses only pre-transition state features: previous proxy reward, judge scores,

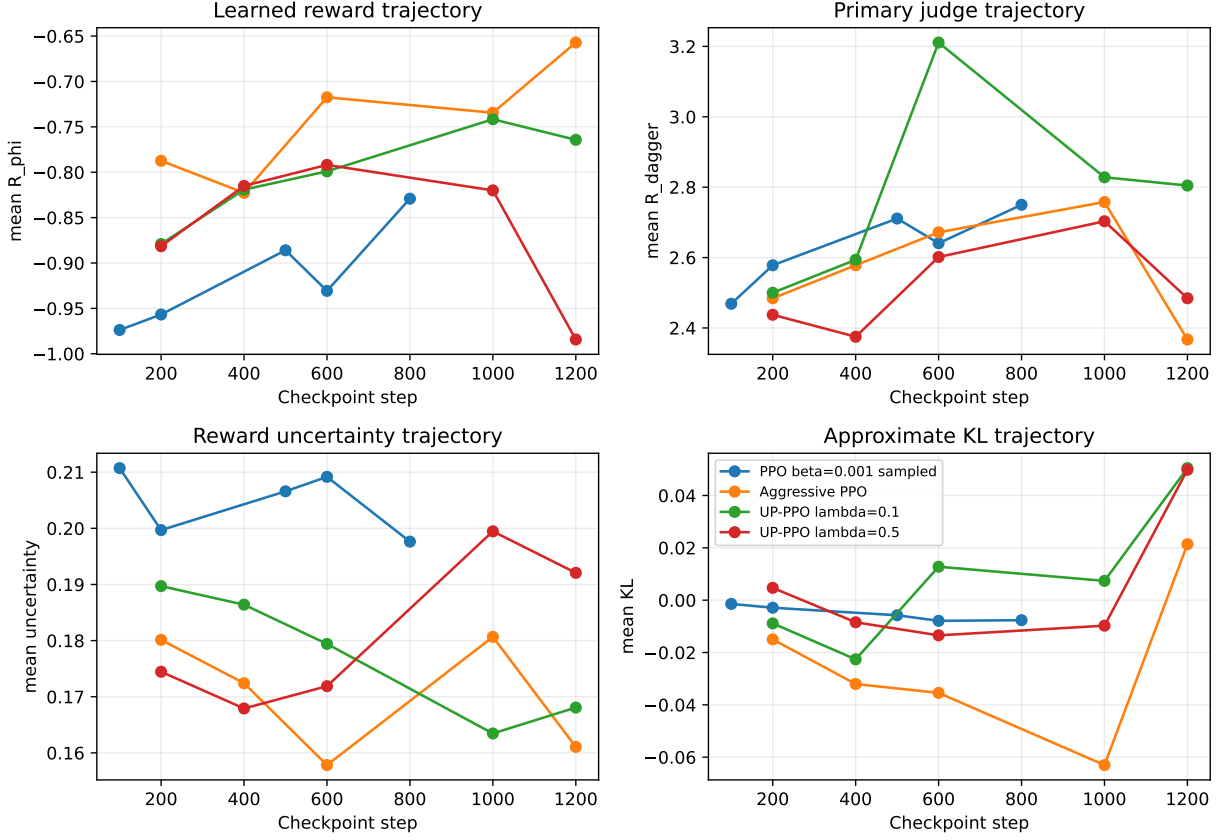


Figure 2: Temporal trajectories for PPO-family runs. The panels track mean learned reward, primary judge score, reward uncertainty, and approximate KL across evaluated checkpoints.

uncertainty, KL, judge disagreement, length, diversity, and repetition. This model reaches ROC-AUC of 0.821 with logistic regression; a random-forest robustness check gives ROC-AUC of 0.813. Average precision is lower (0.256 and 0.209, respectively) because reward hacking is rare in the test set (6.6% prevalence), but both models are substantially above prevalence.

Transition-diagnostic models using ΔR_ϕ and $\Delta R^\dagger$ reach near-perfect performance, as expected, because the label is defined from the signs of these deltas. We treat these rows as a sanity check and emphasize the pre-state-only result as the meaningful early-warning evidence.

5.7 Early-warning feature importance

Table 6 reports the strongest features from the early-warning models. We separate two settings. The first, pre-state-only, uses information available before the transition: previous reward-model score, judge scores, uncertainty, KL, judge disagreement, length, diversity, and repetition. This is the meaningful early-warning setting because it does not observe the future deltas that define the label. The second, transition diagnostics, includes ΔR_ϕ , $\Delta R^\dagger$, and related post-transition quantities. Those features serve as a sanity check because the reward-hacking label is itself defined by the signs of proxy and judge changes.

For the pre-state-only logistic model, previous primary judge score has the largest standardized coefficient. Response length and distinct-1 diversity also appear among the largest positive coefficients, while previous proxy reward, repetition, distinct-2 diversity, KL, judge disagreement, uncertainty, and the second judge score have negative coefficients in this fitted model. These signs should not be interpreted causally.

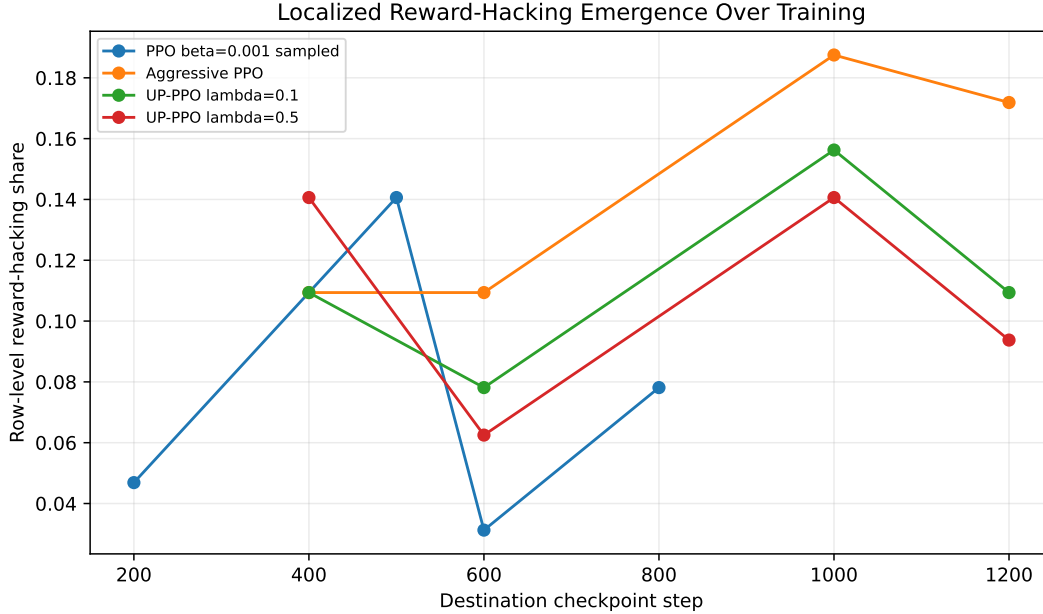


Figure 3: Localized reward-hacking emergence over training, measured as the row-level reward-hacking share at the destination checkpoint of each transition.

Table 5: Early-warning prediction of future row-level reward hacking. Pre-state-only features are measured before the transition and are therefore the relevant warning result.

Feature set	Model	ROC-AUC	Avg. precision	Precision	Recall	F1
Pre-state only	Logistic regression	0.821	0.256	0.167	0.789	0.275
Pre-state only	Random forest	0.813	0.209	0.235	0.105	0.145
Transition diagnostics	Logistic regression	0.974	0.606	0.458	1.000	0.628
Transition diagnostics	Random forest	1.000	1.000	1.000	1.000	1.000

They indicate how the classifier separates future reward-hacking rows within this artifact collection after standardization and correlation among features. The random forest distributes importance more evenly, with previous judge score, previous KL, previous proxy reward, length, diversity, and uncertainty all contributing. This pattern supports the paper’s more cautious claim: future localized reward hacking is partially predictable from pre-transition state, but no single diagnostic is sufficient.

The practical interpretation is triage, not certification. A pre-state-only ROC-AUC near 0.82 indicates that the warning features contain signal, but the rarity and localization of reward hacking mean that precision remains limited. In deployment, such a model would be most useful for prioritizing prompts, checkpoints, or training intervals for additional judge calls or human review.

5.8 Qualitative transition examples

The quantitative taxonomy is reflected in individual generations. Appendix 8 gives representative matched prompt transitions, including before/after response snippets, proxy scores, judge scores, and assigned modes. In a tax-advice prompt, for example, the learned reward increases while both judges fall sharply as the response becomes less coherent and introduces implausible product names. In workplace-bias and privacy-address prompts, the transitions show two qualitatively different failures: strict reward hacking in one case and evaluator gaming in another, where the judges move in opposite